Training-based versus training-free differential privacy for data synthesis

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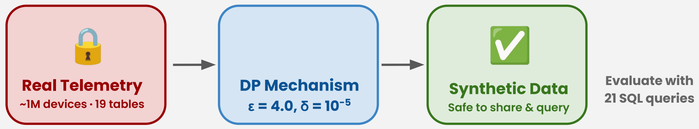
INTRODUCTION

Data drives analytics across healthcare, finance, and device systems. Privacy regulations limit how real-world datasets can be shared. **Differentially private synthetic data** enables analysis while protecting individuals

DP-SGD (training-based) **Private Evolution (training-free)**



DATA & PROBLEM

Intel DCA telemetry from **~1M** Windows devices. **19** tables joined by device ID covering hardware, battery, network etc.

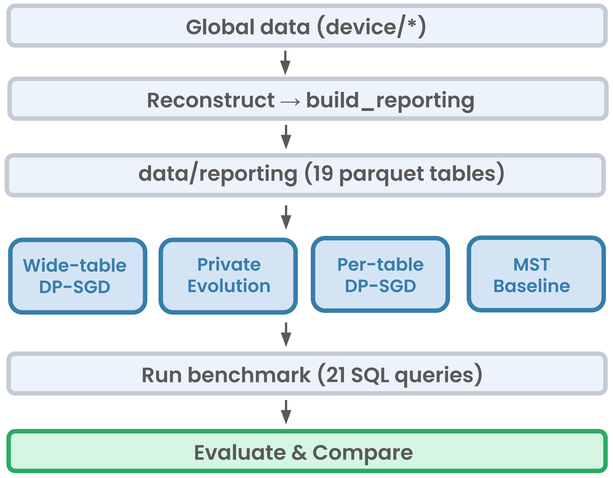
Goal: Generate synthetic data preserving analytical utility while protecting privacy

Metrics: Relative error · Spearman ρ · Total variation. Pass if score **≥ 0.5**

METHODS

DP-SGD (Training-Based)
- Wide table (1 row/device), 70 features
- DP-VAE via Opacus: gradient clipping + Gaussian noise
- 1M synthetic records -> reporting tables

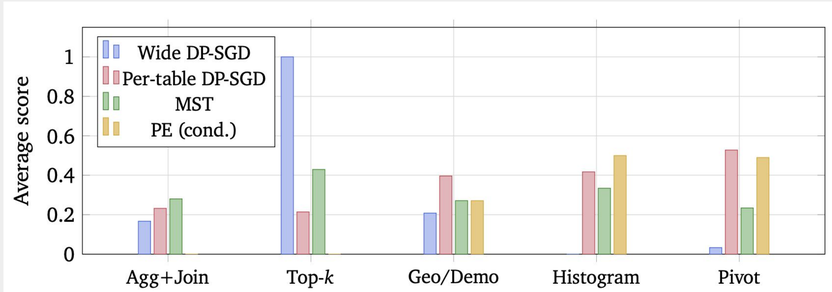
Private Evolution (Training-free)
- LLM generates candidate records
- Real devices vote via workload-aware distance + noise
- Top candidates selected as synthetic data

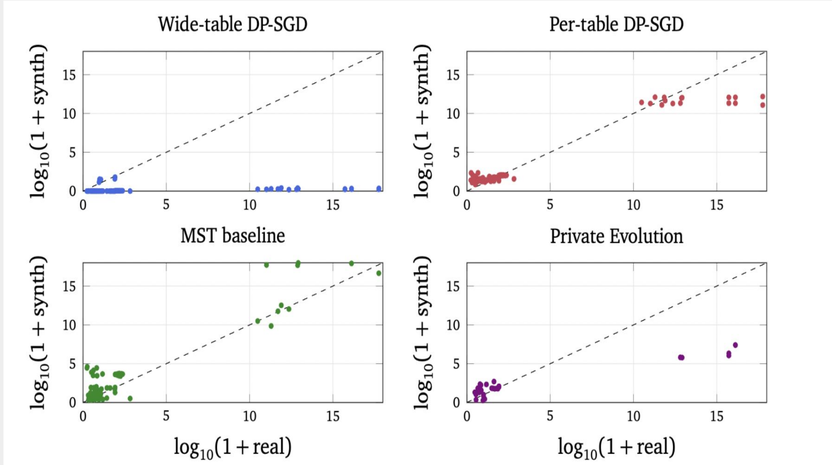


RESULTS

Query	Type	Wide	Per-table	MST	PE (cond.)
BlockerAgg	Agg+Join	—	0.250	0.250	—
BlockerDurAgg	Agg+Join	—	0.000	0.000	—
ChassisAgg	Agg+Join	0.167	0.000	0.000	0.000
DisplayAgg	Agg+Join	—	0.000	0.000	—
NetAsymAgg	Agg+Join	—	0.143	0.429	—
VendorAgg	Agg+Join	—	1.000	1.000	—
BatteryDemo	Geo/Demo	0.000	0.000	0.000	0.333
BatteryGeo	Geo/Demo	0.250	1.000	0.500	0.250
PowerGeo	Geo/Demo	0.333	0.333	0.333	0.000
XeonGeo	Geo/Demo	0.250	0.250	0.250	0.500
BrowserHist	Histogram	0.000	0.333	0.667	0.000
RamHist	Histogram	0.000	0.500	0.000	1.000
PersonaPivot	Pivot	0.065	0.419	0.194	0.161
SleepPivot	Pivot	0.000	0.636	0.273	0.818
BrowserRank	Top-*k*	1.000	1.000	1.000	0.000
DetectionRank	Top-*k*	—	0.500	1.000	—
FocalRank	Top-*k*	—	0.000	0.000	—
SystemRank	Top-*k*	—	0.000	1.000	—
WaitModeRank	Top-*k*	—	0.000	0.000	—
WaitRank	Top-*k*	—	0.000	0.000	—
WaitStateRank	Top-*k*	—	0.000	0.000	—

Per-table DP-SGD	Wide-table DP-VAE	Private Evolution	MST Baseline
6/21	1/10	3/10	6/21





DISCUSSION

Joins fail: Per-table synthesis preserves within table structure but **breaks cross-table correlations**

Wide-table fails: it preserves shared structure but fails under extreme sparsity

PE is mixed: improves when conditioned on DP aggregates, but LLM introduce distribution errors.

CONCLUSION

Results: Per-table DP-SGD and MST: 6/21. Wide DP-VAE: 1/10. PE (cond.) : 3/10.

Takeaway: Join-heavy aggregates are hardest, wide-table collapses under sparsity, per-table breaks cross-table relationships.

Next Steps: Relational DP synthesis is the natural next step.